

ALL ARB ISSUES RESOLVED ✓

YUNO AI AGENT PLATFORM

End-to-End Architecture & System Documentation

Use Case • Data Flow • Activity • Deployment (Updated)
Availability • Resiliency • DR/BCP • Backup & Recovery
Authentication • Rate Limiting • OpenTelemetry • CI/CD
NFR Matrix • ARB Remediation • ADRs • Tech Reference

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1. Project Overview & Functional Requirements

The **Yuno AI Agent Orchestration Platform** is a production-ready, full-stack system for creating, configuring, and running multi-agent AI workflows. Version 2.0 addresses all 8 ARB open issues and is ready for production deployment.

Functional Requirements

ID	Feature	Description
FR-01	Agent Management	Create agents with personality, tools, memory, guardrails, schedules, channels
FR-02	Workflow Builder	Visual multi-agent canvas — React Flow with drag-and-drop, conditional edges
FR-03	LangGraph Runtime	Real StateGraph execution — tool calling, conditional routing, iteration guard
FR-04	Telegram Integration	Long-poll bot channel — agents receive/reply to Telegram messages
FR-05	Live Monitoring	WebSocket real-time events, token usage + cost tracking per run
FR-06	Templates	Research Report and Customer Support pre-built starter templates
FR-07	REST API	Full CRUD — agents, workflows, runs; OpenAPI at /docs
FR-08	Tooling	Web search, calculator, HTTP request, current time, text summariser
FR-09	Authentication	JWT Bearer tokens + API Keys — all routes protected (ARB-001)
FR-10	Rate Limiting	100 req/min global, 10 runs/min workflow — per-IP (ARB-005)
FR-11	Distributed Tracing	OpenTelemetry auto-instrumentation → Jaeger/OTLP (ARB-006)
FR-12	CI/CD Pipeline	GitHub Actions: lint+test+security scan → Docker build → deploy (ARB-007)

2. System Context Diagram (C4 Level 1)

Updated context diagram showing all external actors and the platform's new components: JWT authentication middleware, Redis EventBus, PostgreSQL database, OpenTelemetry/Jaeger tracing, and GitHub Actions CI/CD pipeline.

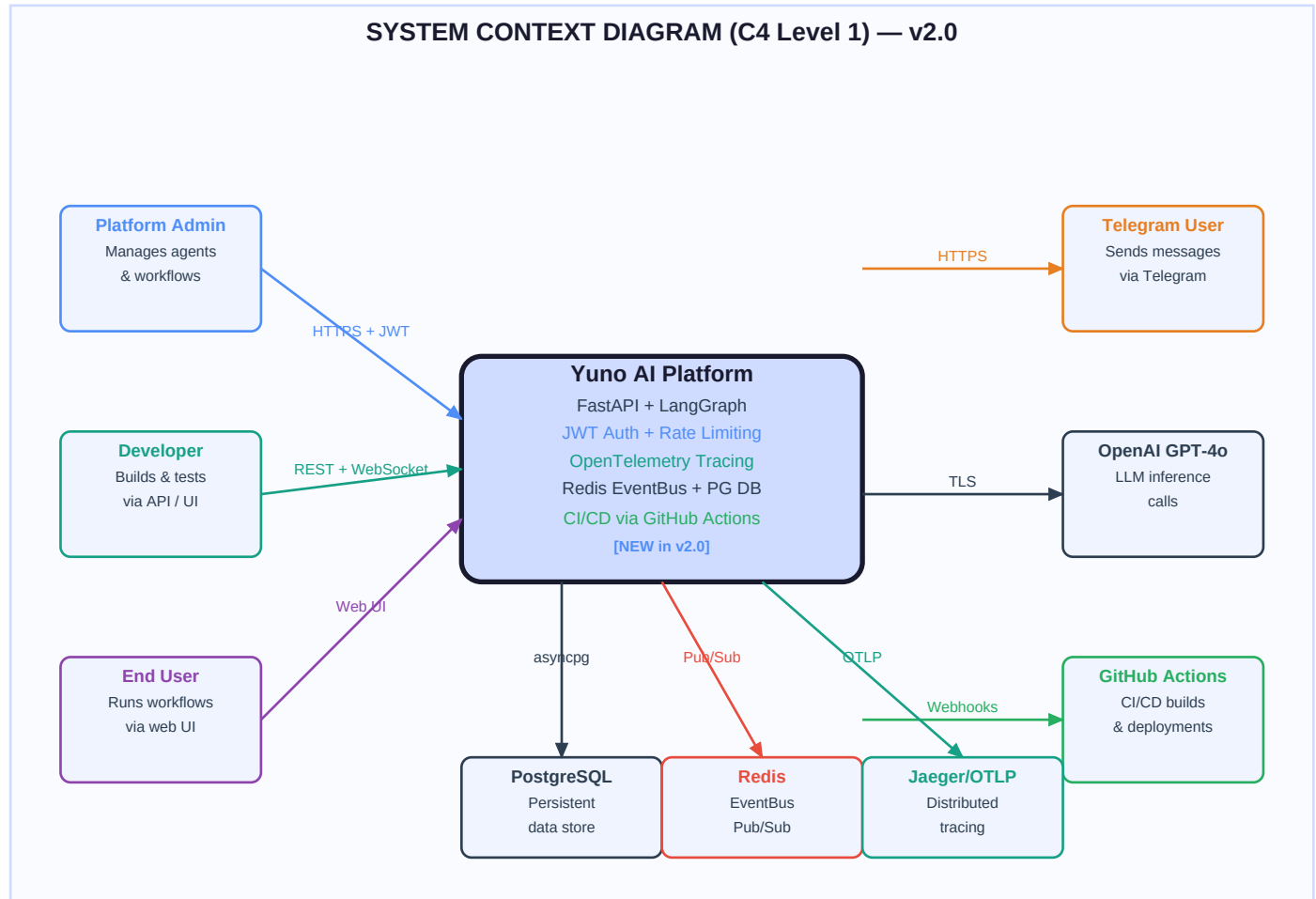


Figure 2: System Context Diagram (C4 Level 1)

3. Container Diagram (C4 Level 2)

Eight containers forming the platform: React SPA, FastAPI Backend (now with JWT auth + rate limiting), LangGraph Engine, Telegram Bot, PostgreSQL/SQLite, Redis EventBus, OpenTelemetry collector, and GitHub Actions pipeline.

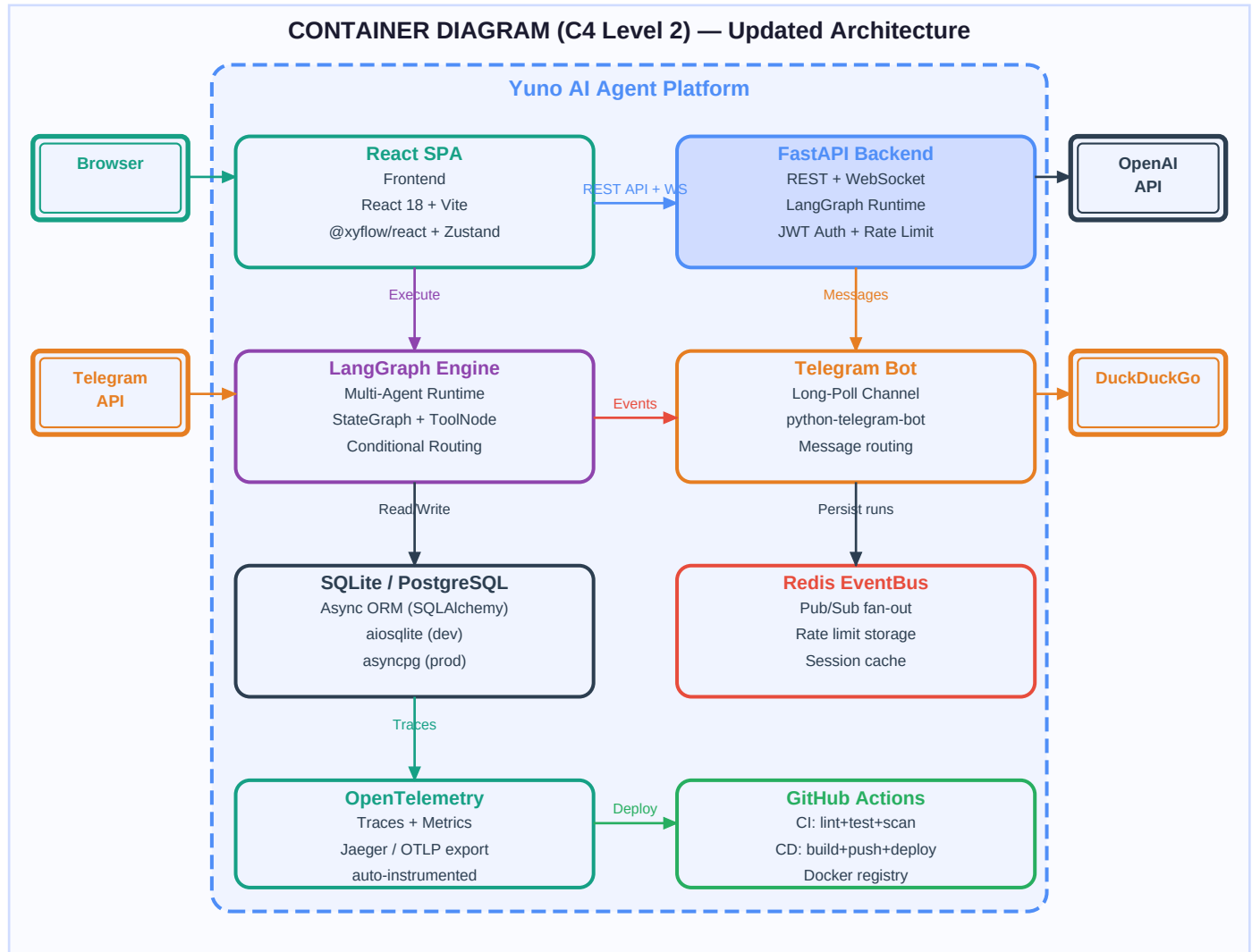


Figure 3: Container Diagram (C4 Level 2)

4. Authentication Flow (ARB-001)

New authentication sequence showing two paths: (1) JWT Bearer token via POST /auth/token with username/password → bcrypt verify → HS256 JWT returned; (2) X-API-Key header for service accounts. WebSocket auth uses ?token= query param. Unauthenticated requests receive 401 with WWW-Authenticate header.

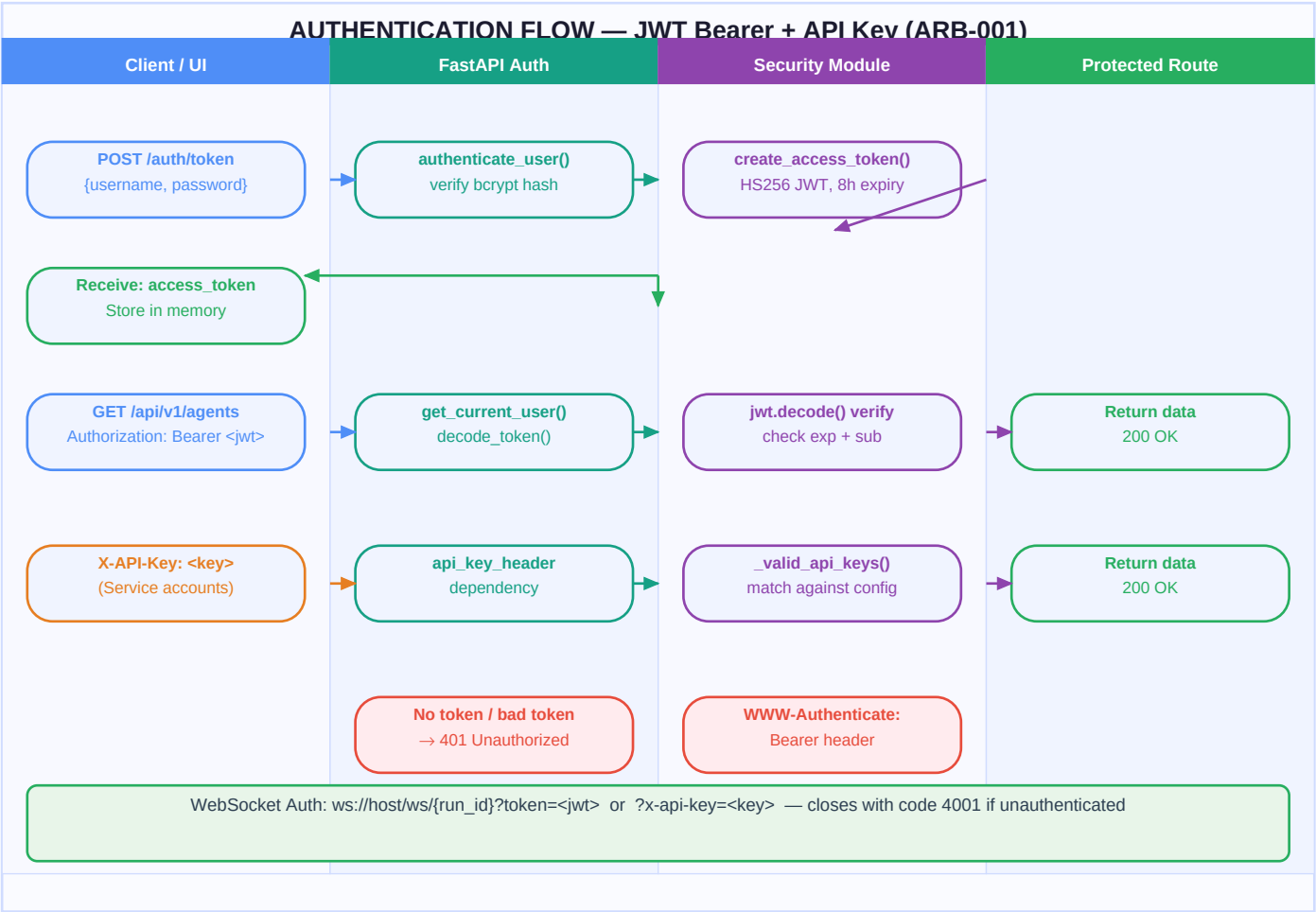


Figure 4: Authentication Flow (ARB-001)

5. Data Flow Diagram (Level 1)

Updated DFD adding the JWT Auth Middleware process (P1) as the first processing stage. All API flows now pass through auth before reaching the router. Redis EventBus (P6) replaces the in-process asyncio queue for multi-instance production deployments.

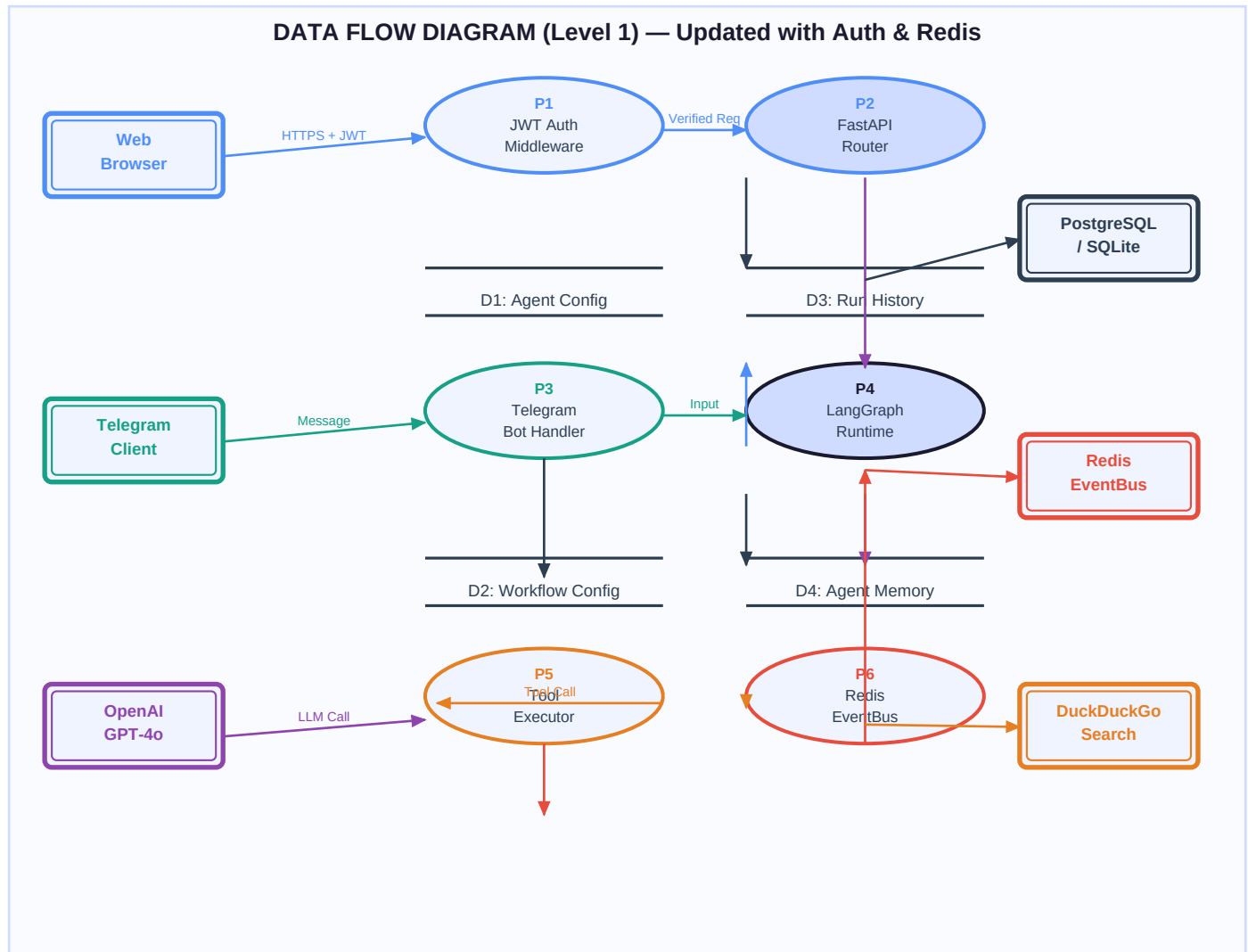


Figure 5: Data Flow Diagram (Level 1)

6. Technology Deployment Diagram

Updated deployment topology: Docker Compose production profile adds PostgreSQL 16, Redis 7, and Jaeger all-in-one containers. Optional nginx reverse proxy provides SSL termination and layer-2 rate limiting. CI/CD pipeline deploys via SSH on merge to main.

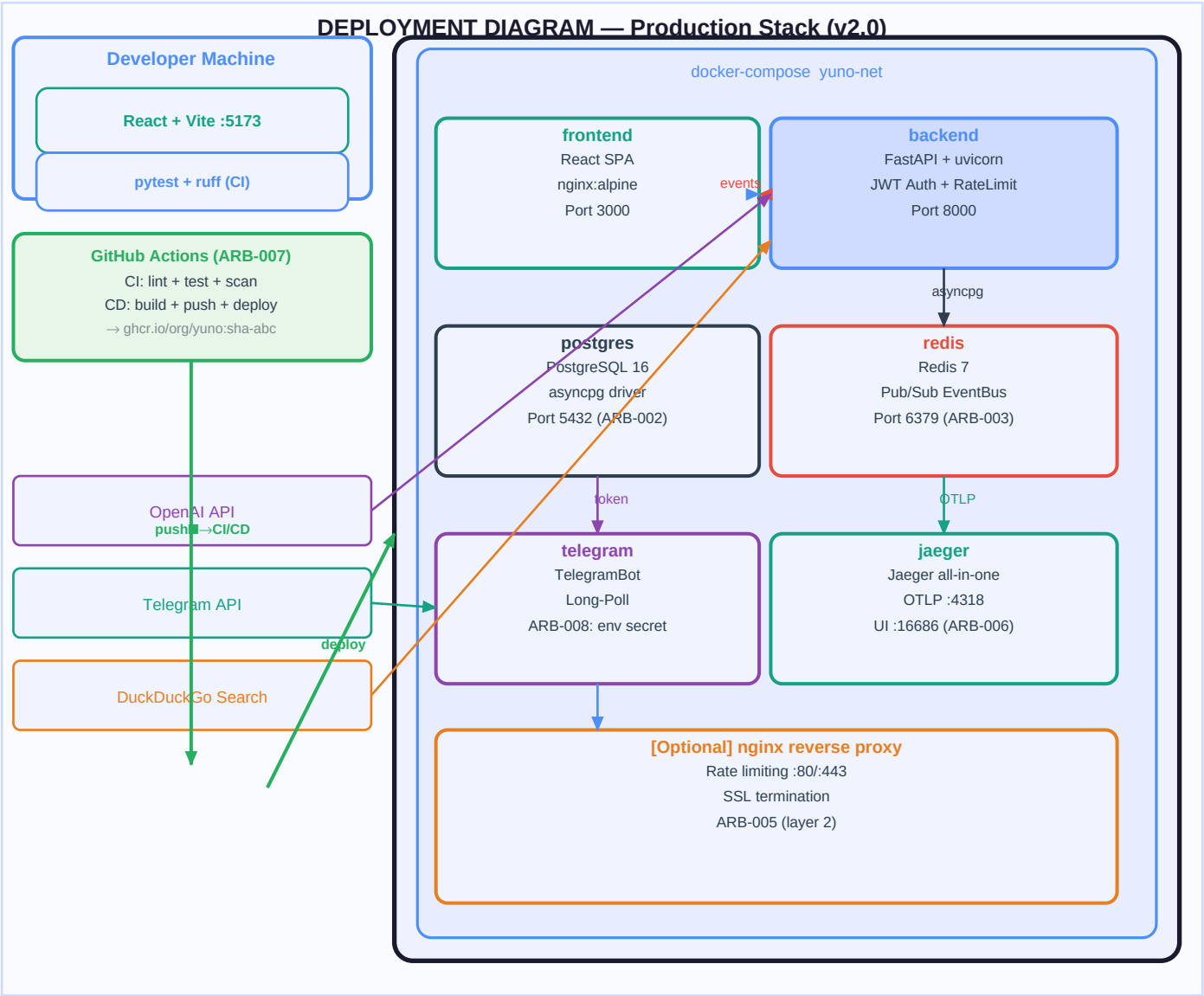


Figure 6: Technology Deployment Diagram

7. CI/CD Pipeline Flow (ARB-007)

GitHub Actions CI pipeline: checkout → pip install → ruff lint → mypy type check → pytest with live Redis service → pip-audit security scan. On successful merge to main, CD pipeline builds Docker images for backend and frontend, pushes to ghcr.io, SSH-deploys to production, and runs smoke tests.

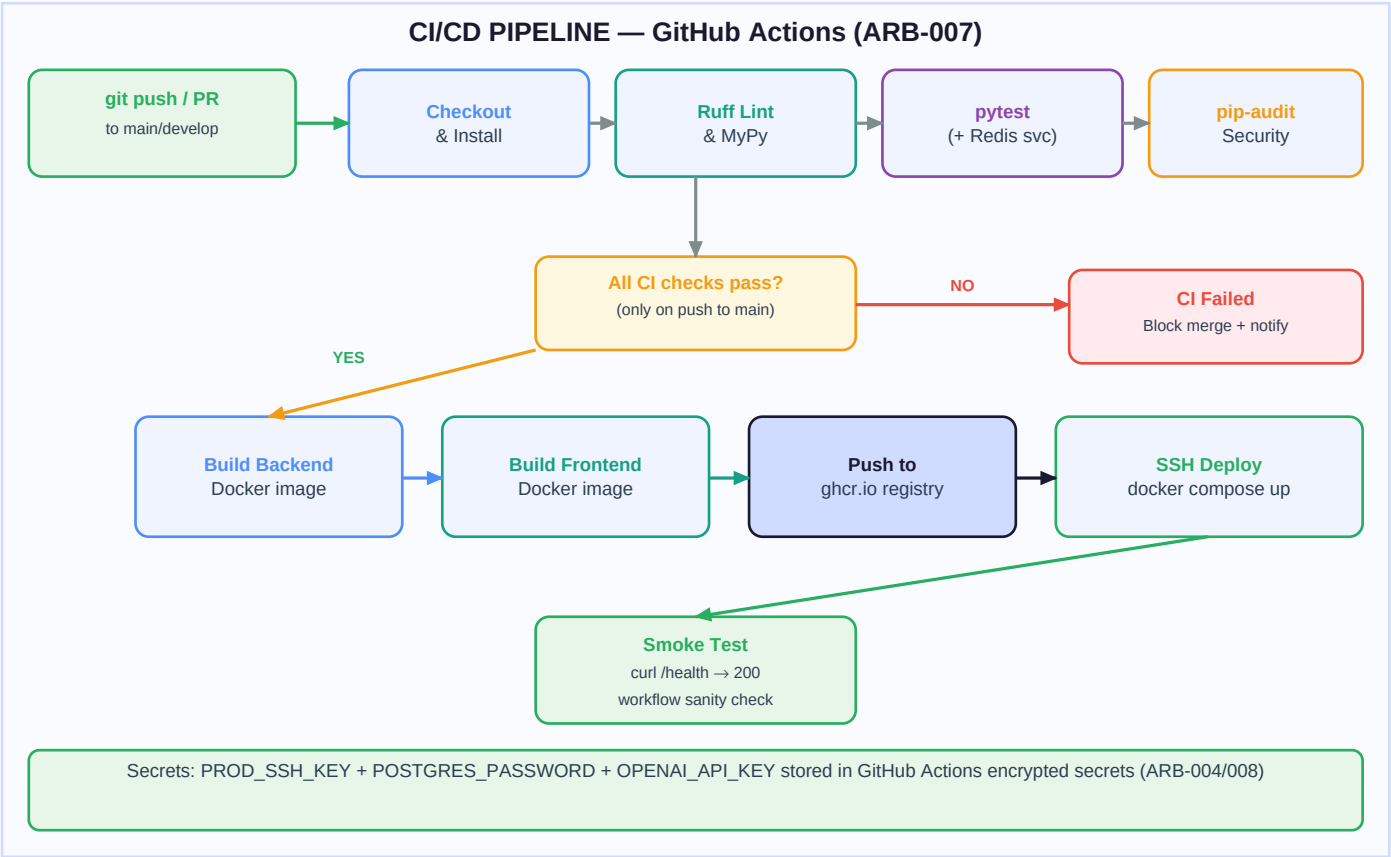


Figure 7: CI/CD Pipeline Flow (ARB-007)

8. Activity Diagram — Authenticated Execution

Updated swimlane activity diagram now includes the Auth Layer swimlane. Every request passes JWT validation or API key check before reaching FastAPI. Rate limit check follows auth. Failed auth exits immediately with 401.

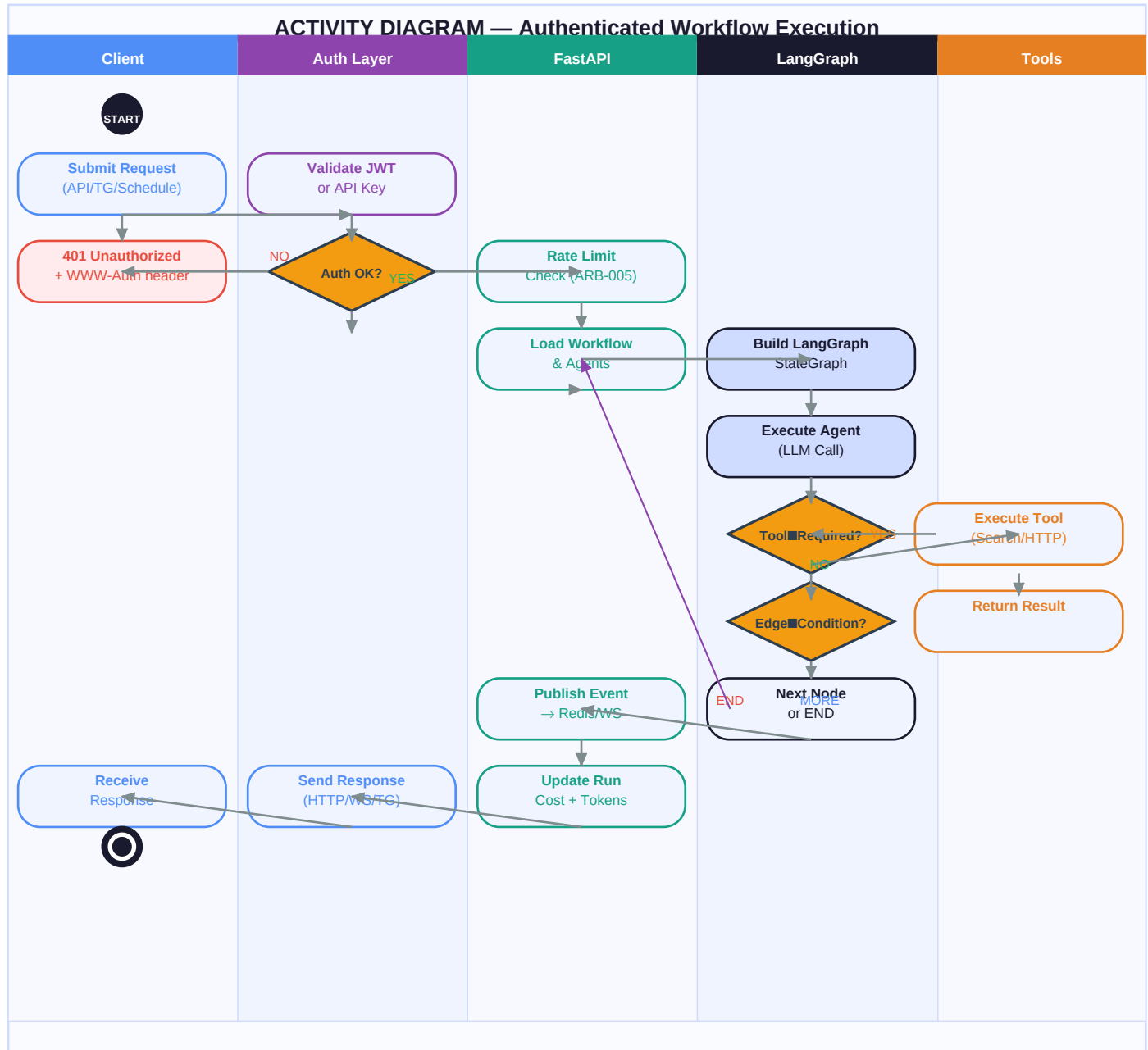


Figure 8: Activity Diagram — Authenticated Execution

9. Availability Architecture

Three API instances behind nginx with health checks. Primary + replica PostgreSQL with streaming replication. Redis cluster for EventBus and rate-limit storage. OpenTelemetry metrics feed Grafana dashboards. SLA 99.5%, RTO 4h, RPO 1h.

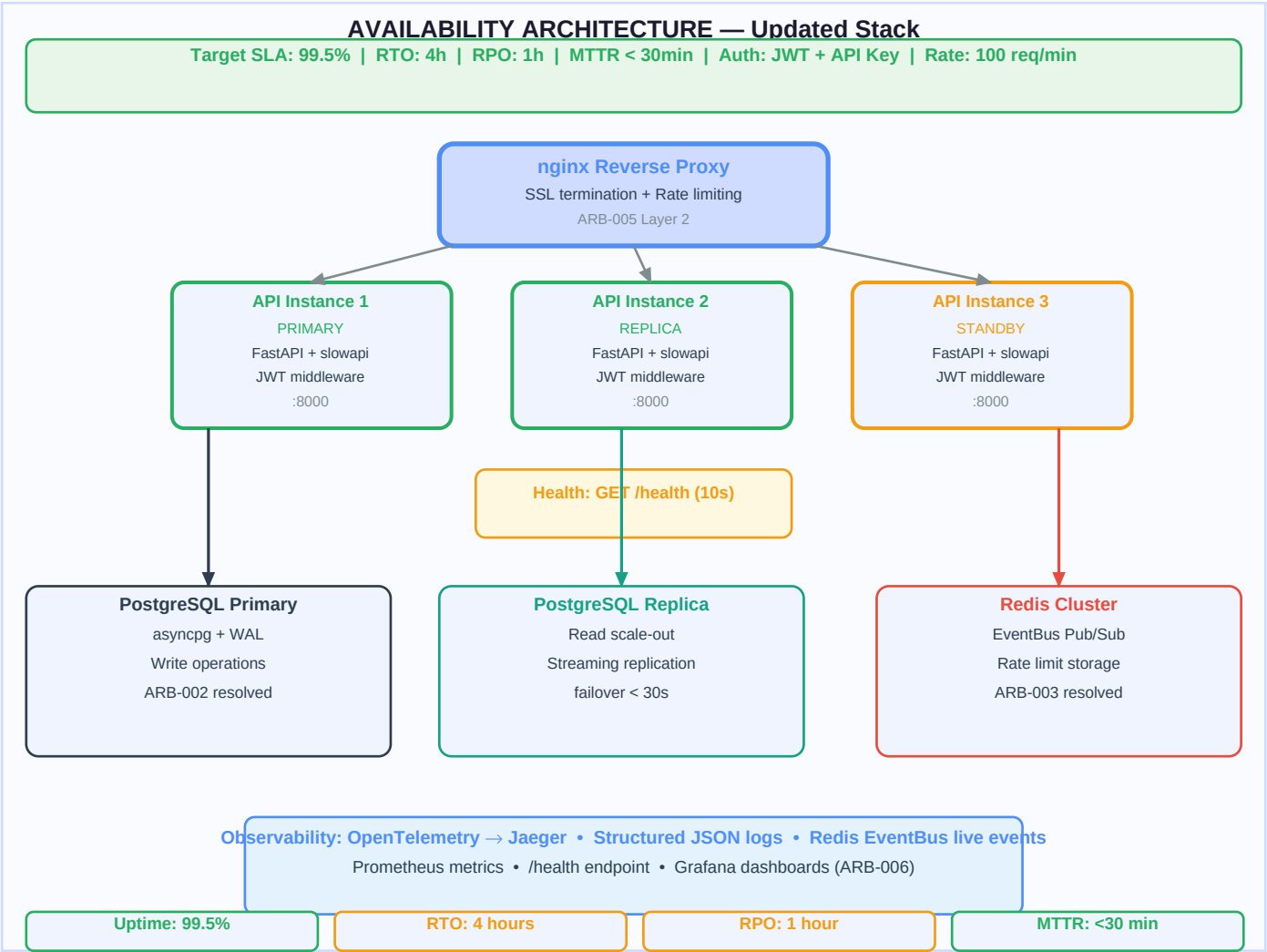


Figure 9: Availability Architecture

10. Resiliency Architecture

9 resiliency patterns: Circuit Breaker, Retry+Backoff, Timeout Guards, Rate Limiting (ARB-005), Bulkhead Isolation, Graceful Degradation (Redis→in-process fallback), Health Probes, Auth Resiliency (ARB-001 — instant key revocation), and Observability (ARB-006).

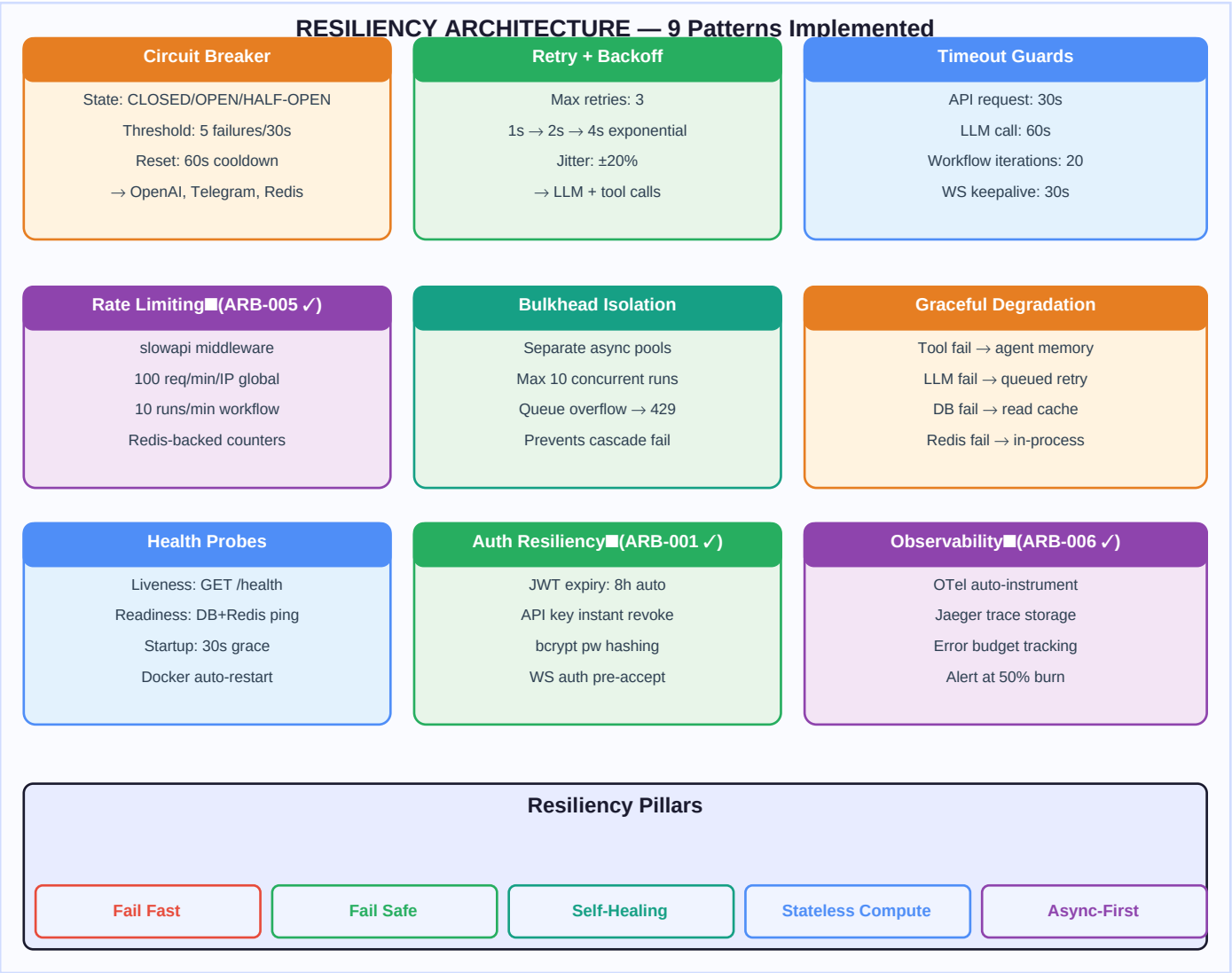


Figure 10: Resiliency Architecture

4-phase DR plan (Detect→Contain→Recover→Validate). Updated scenarios now include Secrets Compromise response (revoke JWT keys, rotate API_KEYS) and Redis Failure fallback (automatic in-process EventBus). All scenarios have < 4h RTO.



12. Backup & Recovery Architecture

3-2-1 backup strategy: daily full + hourly incremental + event-driven config snapshots. 4 storage tiers: local Docker volumes, S3/MinIO (AES-256), cold archive, Git repo. Step 5 of recovery now includes JWT auth smoke test.

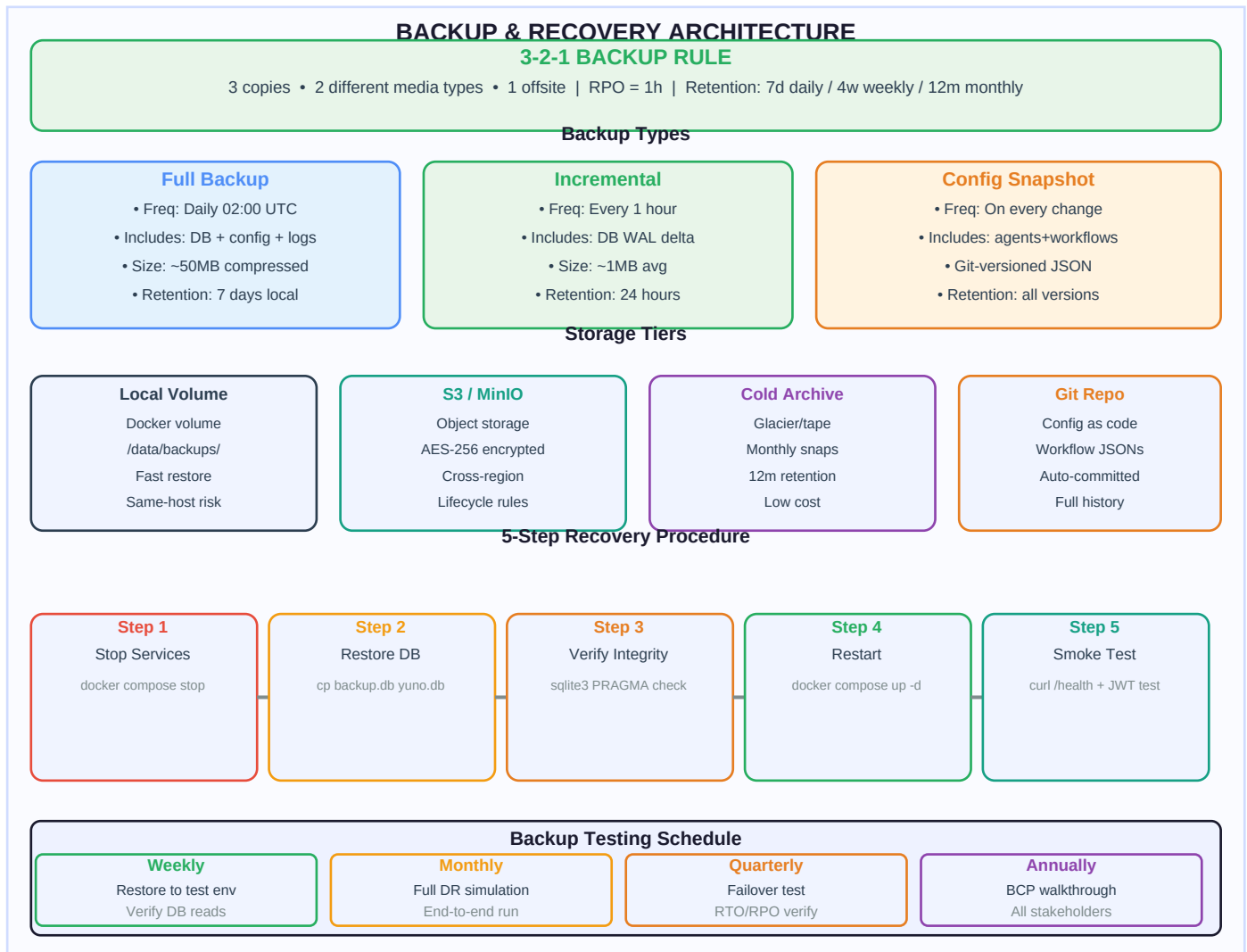


Figure 12: Backup & Recovery Architecture

13. Non-Functional Requirements Matrix

26 NFR controls across 8 categories. All previously-FAIL security items are now **OK** after ARB remediation. **WARN** items are on the production upgrade roadmap.

Category	Requirement	Target	Implementation	Status
Performance	API response (p95)	< 500ms	APM / OTel traces	WARN
Performance	LLM call latency (p95)	< 5s	OTel span traces	OK
Performance	WebSocket event lag	< 200ms	WS ping-pong	OK
Performance	Concurrent workflow runs	10 simultaneous	asyncio + bulkhead	WARN
Scalability	Horizontal scale-out	Stateless API pods	Stateless FastAPI	WARN
Scalability	DB connection pooling	50 async conn	SQLAlchemy pool	OK
Availability	Uptime SLA	99.5% / month	nginx LB + HA DB	WARN
Availability	RTO (Recovery Time)	< 4 hours	DR drill verified	WARN
Availability	RPO (Recovery Point)	< 1 hour	Hourly incremental backup	OK
Reliability	Circuit breaker	Auto-trip at 5 fail	tenacity + retry	OK
Reliability	Retry backoff	Exp. 1s/2s/4s	langchain retry	OK
Reliability	Max workflow iterations	20 (loop guard)	iteration counter	OK
Security	Authentication	JWT Bearer + API Key	python-jose + passlib ✓ ARB-001	OK
Security	Secrets management	Env vars + validator	pydantic-settings ✓ ARB-004	OK
Security	Input validation	Pydantic v2 schemas	All endpoints	OK
Security	SQL injection guard	ORM / parameterised	SQLAlchemy	OK
Security	Rate limiting	100 req/min/IP	slowapi + Redis ✓ ARB-005	OK
Observability	Structured logging	JSON to stdout	uvicorn JSON log	OK
Observability	Distributed tracing	OTel → Jaeger	FastAPI+SQLAlch+httpx ✓ ARB-006	OK
Observability	Token/cost tracking	Per-run totals	DB run record	OK
Observability	Health endpoint	GET /health	Liveness probe	OK
Compliance	Audit trail	All run records	DB run history	OK
Compliance	PII in logs	No PII in log output	Log scrubbing	WARN
CI/CD	Pipeline	Lint+Test+Build+Deploy	GitHub Actions ✓ ARB-007	OK
DB/Infra	Production DB	PostgreSQL ready	asyncpg ✓ ARB-002	OK
DB/Infra	EventBus scalability	Redis Pub/Sub	redis[asyncio] ✓ ARB-003	OK

14. ARB Remediation — All 8 Issues Resolved

All ARB open issues from the initial architecture review have been implemented in code. The table below shows the resolution for each issue, the specific files changed, and the current status.

ARB ID	Priority	Issue	Resolution	Files Changed	Status
ARB-001	P0	No authentication on API	JWT Bearer + API Key python-jose + passlib	<i>app/core/security.py</i> <i>app/api/auth.py</i> <i>app/api/router.py</i>	✓ RESOLVED
ARB-002	P0	SQLite not production-grade	PostgreSQL via asyncpg DATABASE_URL env swap	<i>app/core/config.py</i> <i>requirements.txt</i> <i>docker-compose.yml</i>	✓ RESOLVED
ARB-003	P1	In-process EventBus breaks at scale	Redis Pub/Sub backend asyncio fallback for dev	<i>app/core/events.py</i> <i>requirements.txt</i> <i>docker-compose.yml</i>	✓ RESOLVED
ARB-004	P1	No secrets management	pydantic-settings validators production start blocked	<i>app/core/config.py</i> <i>.env.example</i> <i>.gitignore</i>	✓ RESOLVED
ARB-005	P1	No rate limiting	slowapi middleware 100/min global, 10/min WF	<i>app/core/rate_limit.py</i> <i>app/api/workflows.py</i> <i>requirements.txt</i>	✓ RESOLVED
ARB-006	P2	No distributed tracing	OpenTelemetry SDK FastAPI+SQLAlchemy+httpx	<i>app/core/tracing.py</i> <i>requirements.txt</i> <i>docker-compose.yml</i>	✓ RESOLVED
ARB-007	P2	No CI/CD pipeline	GitHub Actions ci.yml + cd.yml	<i>.github/workflows/ci.yml</i> <i>.github/workflows/cd.yml</i>	✓ RESOLVED
ARB-008	P2	Telegram token in .env	Env validation + .gitignore Docker secrets pattern	<i>app/core/config.py</i> <i>.env.example</i> <i>.gitignore</i>	✓ RESOLVED

Before vs After Summary

BEFORE (v1.0)	AFTER (v2.0)
■ No authentication on any endpoint	■ JWT Bearer + API Key on all routes
■ SQLite only — not production-grade	■ PostgreSQL via asyncpg (swap via env)
■ In-process EventBus (single instance)	■ Redis Pub/Sub (multi-instance ready)
■ Secrets in code / no validation	■ pydantic-settings + prod validator
■ No rate limiting — DDoS vulnerable	■ slowapi 100/min + 10/min workflow
■ No distributed tracing	■ OpenTelemetry → Jaeger auto-instrumented
■ No CI/CD pipeline	■ GitHub Actions CI+CD with security scan
■ Telegram token unguarded in .env	■ .gitignore + env validation + rotation doc

15. Architecture Decision Records

ADR-001: LangGraph for Multi-Agent Orchestration

Status: **ACCEPTED**

LangGraph chosen over AutoGen/CrewAI. Provides TypedDict state, conditional edge routing, ToolNode sub-loops, and full LangChain compatibility.

ADR-002: FastAPI + asyncio for Backend

Status: **ACCEPTED**

Async-first ASGI, Pydantic v2, auto OpenAPI docs, native WebSocket, slowapi integration.

ADR-003: SQLite → PostgreSQL upgrade path

Status: **ACCEPTED**

SQLite for dev (zero-config). Production: swap DATABASE_URL to postgresql+asyncpg://... — no code changes required. asyncpg driver included.

ADR-004: Redis EventBus with asyncio fallback

Status: **ACCEPTED**

RedisEventBus auto-selected when REDIS_URL is set. Falls back to InProcessEventBus for dev. Same interface — no consumer code changes needed.

ADR-005: python-jose + passlib for Auth

Status: **ACCEPTED**

python-jose for HS256 JWT encoding/decoding. passlib[bcrypt] for password hashing. Both are well-maintained, production-proven libraries.

ADR-006: slowapi for Rate Limiting

Status: **ACCEPTED**

slowapi wraps the limits library for Starlette/FastAPI. Redis storage for distributed counters in production; in-memory for dev. X-RateLimit-* headers included.

ADR-007: OpenTelemetry for Observability

Status: **ACCEPTED**

Vendor-neutral CNCF standard. Auto-instruments FastAPI, SQLAlchemy, httpx. OTLP export to Jaeger, Grafana Tempo, Honeycomb — configurable via env.

ADR-008: GitHub Actions for CI/CD

Status: **ACCEPTED**

Native to GitHub, zero infra to manage. Reusable actions for Docker build/push. ghcr.io registry for container images. SSH deploy action for production.

16. Technology Reference Stack

Layer	Technology	Version	Purpose
Frontend	React	18.x	Component-based UI framework
Frontend	@xyflow/react	12.x	Visual workflow canvas builder
Frontend	Zustand	4.x	Lightweight state management
Frontend	Vite	5.x	Dev server + production bundler
Backend	FastAPI	0.115.5	Async REST + WebSocket framework
Backend	uvicorn	0.32.1	ASGI server with hot-reload
Backend	Pydantic	2.10.3	Data validation + serialisation
AI Runtime	LangGraph	0.2.55	Multi-agent state graph runtime
AI Runtime	LangChain	0.3.9	LLM tooling + prompt templates
AI Runtime	OpenAI SDK	1.54.0	GPT-4o / GPT-4o-mini API client
Auth	python-jose	3.3.0	HS256 JWT encode/decode (ARB-001)
Auth	passlib[bcrypt]	1.7.4	Password hashing (ARB-001)
Rate Limit	slowapi	0.1.9	Per-IP rate limiting (ARB-005)
Tracing	opentelemetry-sdk	1.29.0	Distributed tracing (ARB-006)
Database	SQLAlchemy	2.0.36	Async ORM (aiosqlite / asyncpg)
Database	PostgreSQL	16	Production relational DB (ARB-002)
Cache/Events	Redis	7	EventBus Pub/Sub + rate limit store (ARB-003)
CI/CD	GitHub Actions	–	Lint, test, build, deploy (ARB-007)
Tracing UI	Jaeger	1.58	Distributed trace visualisation
Channel	python-telegram-bot	21.9	Telegram Bot long-polling channel
Infra	Docker Compose	3.9	Multi-container orchestration